Grace Kim

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Education

University of Waterloo Sept. 2023 – Apr. 2028
Bachelor of Applied Science (BASc) in Mechatronics Engineering *Waterloo, ON*

- Cumulative GPA: 91% (3.92/4.00)
- 4x Dean's Honours List (every academic term)

EPFL (Ecole Polytechnique Fédérale de Lausanne) Feb. 2026 – May 2026
Academic Exchange, Mechanical Engineering *Lausanne, Switzerland*

Experience

Robotics Research Intern Mar. 2026 – Aug. 2026
Learning Algorithms and Systems Laboratory (LASA), EPFL *Lausanne, Switzerland*

- Implemented an admittance controller for a 20-DOF Wuji robotic hand based on a virtual mass-spring-damper model to generate compliant and adaptive motion in response to external force input
- Designed and 3D-printed SolidWorks test fixtures to mount and align force/torque sensors with the Wuji robotic hand for controlled data collection and calibration experiments
- Developed a Dockerized ROS control framework for dual KUKA iiwa robotic arms, integrating ROS1/ROS2 communication, trajectory recording and replay, and simulation for repeatable robotics experiments

Undergraduate Research Assistant May 2025 – Dec. 2025
Neuromechanics and Assistive Robotics Lab, University of Waterloo *Waterloo, ON*

- Collected and preprocessed multimodal gait-cycle data from 10 participants using Vicon motion capture, IMUs, EMG sensors, a treadmill, and Bertec force plates, including filtering, sensor synchronization, and gap filling
- Designed and trained neural network models in Keras/TensorFlow, building a hybrid Conv2D-BiLSTM model, to predict one future gait cycle of joint angle trajectories from synchronized time-series sensor data

Systems Analyst Intern Jan. 2025 – Apr. 2025
Mackenzie Investments *Toronto, ON*

- Assisted the Process Automation team in investigating and resolving production issues in an internal platform
- Collaborated with developers, solution architects, and project managers to create low-level design blueprints for major software application releases, detailing API changes, UI enhancements, and non-functional requirements

Process Engineering Intern May 2024 – Aug. 2024
American Biaxis Inc. *Winnipeg, MB*

- Developed Standard Operating Procedures (SOPs) for a 6.6-meter-wide, 9-station film slitter and a Bonfanti automated bridge crane, documenting safe setup, loading/unloading, and handling of 12-ton plastic film rolls
- Conducted tensile, shrinkage, and thickness tests on new Biaxially Oriented Polyamide (BOPA) films, generating detailed analysis reports to support quality assurance and product development

Projects

Beam Balancing Control System | *SolidWorks, MATLAB, Arduino, Raspberry Pi* Sept. 2026 – Oct. 2026

- Designed a servo-actuated beam balancer in SolidWorks, machined and assembled mechanical components, soldered a custom PCB, and integrated Arduino, Raspberry Pi, phototransistors, and multiplexer circuitry
- Developed and tuned a PID feedback controller using webcam and phototransistor inputs to stabilize ball position

Cochlear Implant Signal Processing | *MATLAB, Digital Signal Processing* Sept. 2025 – Dec. 2025

- Built a MATLAB audio signal processing pipeline using multi-channel bandpass filtering, full-wave rectification, and low-pass envelope extraction for frequency-domain decomposition and analysis of speech signals

Technical Skills

Programming: Python, Java, C, MATLAB, R, SQL
Robotics & Controls: Simulink, ROS1/ROS2, Vicon, OptiTrack, Siemens PLC (Ladder Logic)
Engineering & Tools: Arduino, Raspberry Pi, Linux, SolidWorks, Altium Designer, PLECS, Git/GitHub, Jira